SafeNet
The Foundation of Internet Security

Securing your Internet and Intranet Connection

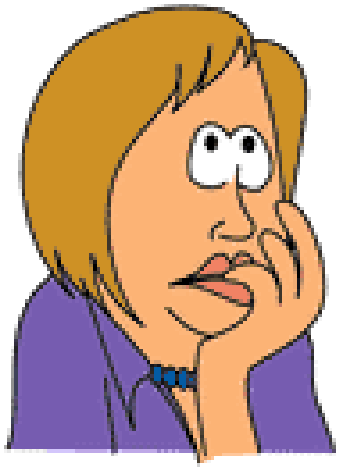


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Abstract

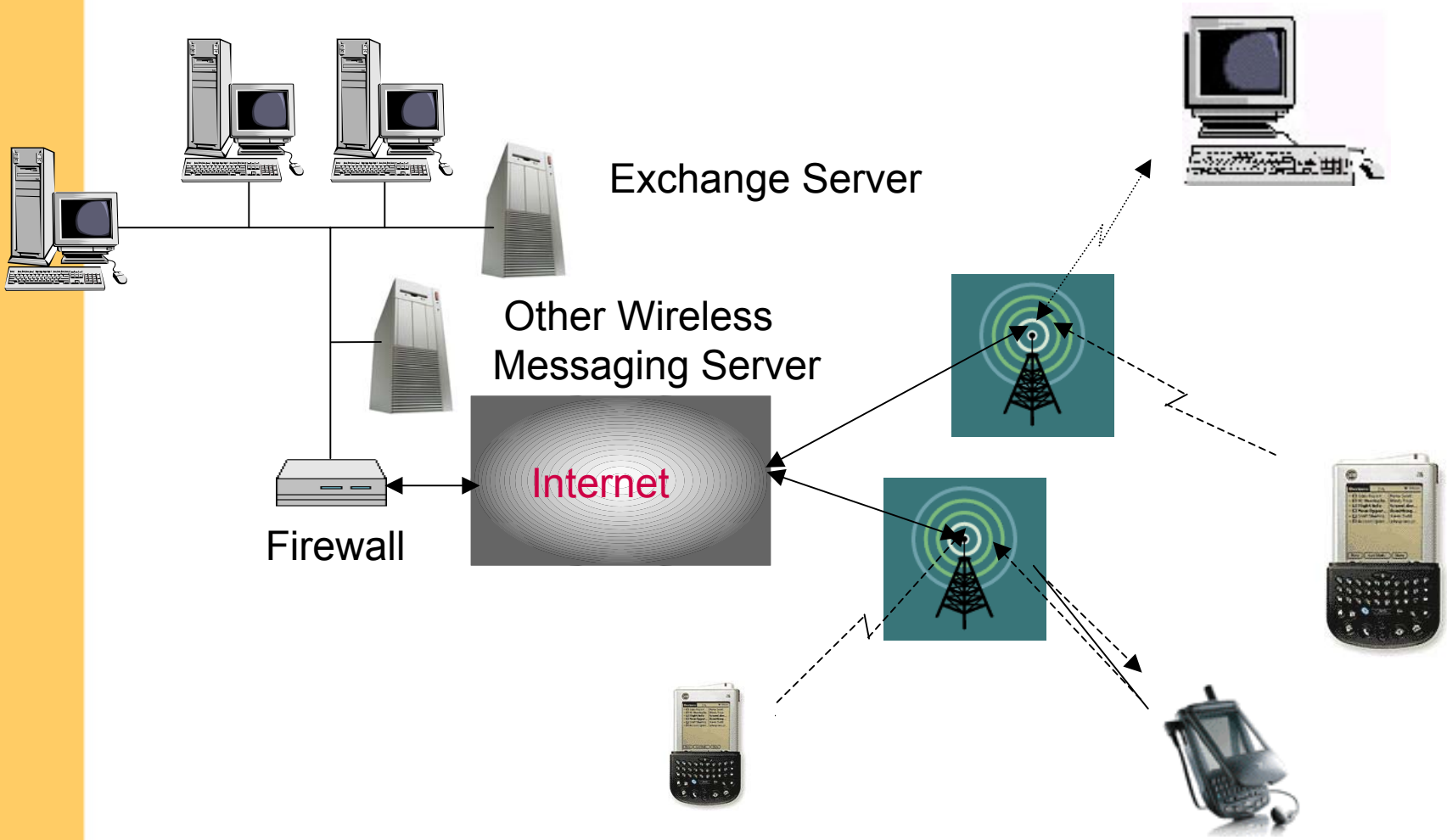
“How can today’s corporate IT manager ensure data security for the tens, hundreds, or even thousands of PDAs now defining the enterprise network perimeter? What safeguards and mechanisms are available (and evolving) to confidently allow secure access to resources within the shop? ”

"Your mission, should you decide to accept it, is to absolutely ensure the security of our corporate network perimeter while hundreds of PDAs of varying makes & models are being deployed to every company employee … tomorrow."

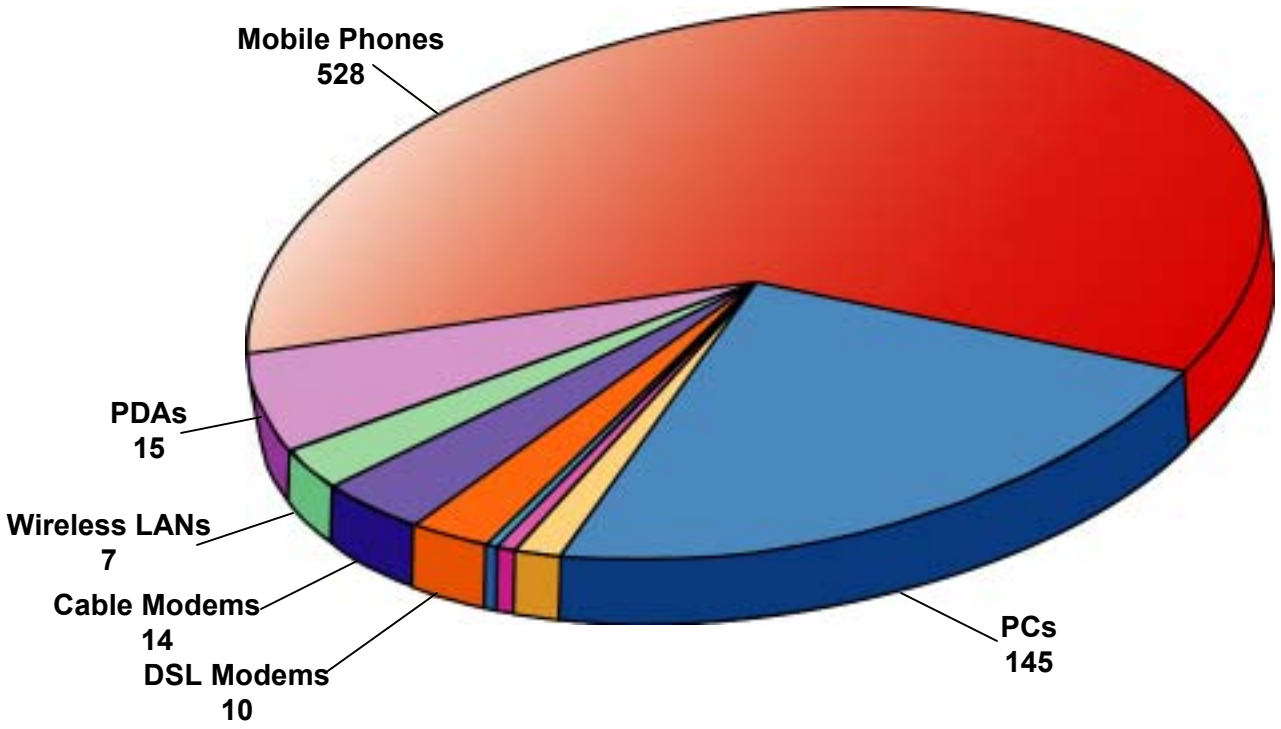




Wireless Phones, PDA's, Laptops are all network devices



The Need for Wireless Security



Total 725.4 Million Units

Security Risks

Physical

- Loss or theft → hacker accesses data files, keys

Visual

- Nosey neighbors on planes, trains & automobiles

Electronic

- Packet sniffers on Internet
- Weak or no encryption
- Weak or no authentication (simple passwords)
- 802.11 sniffer out in corporate office parking lot

What's an IT Manager to do?

Need to provide access to …

- **E-mail servers**
- **Web servers**
- **Network directory servers**
- **Print services**
- **Specialized applications**

✓ ***Policy***
✓ ***Technology***
✓ ***Procedure***

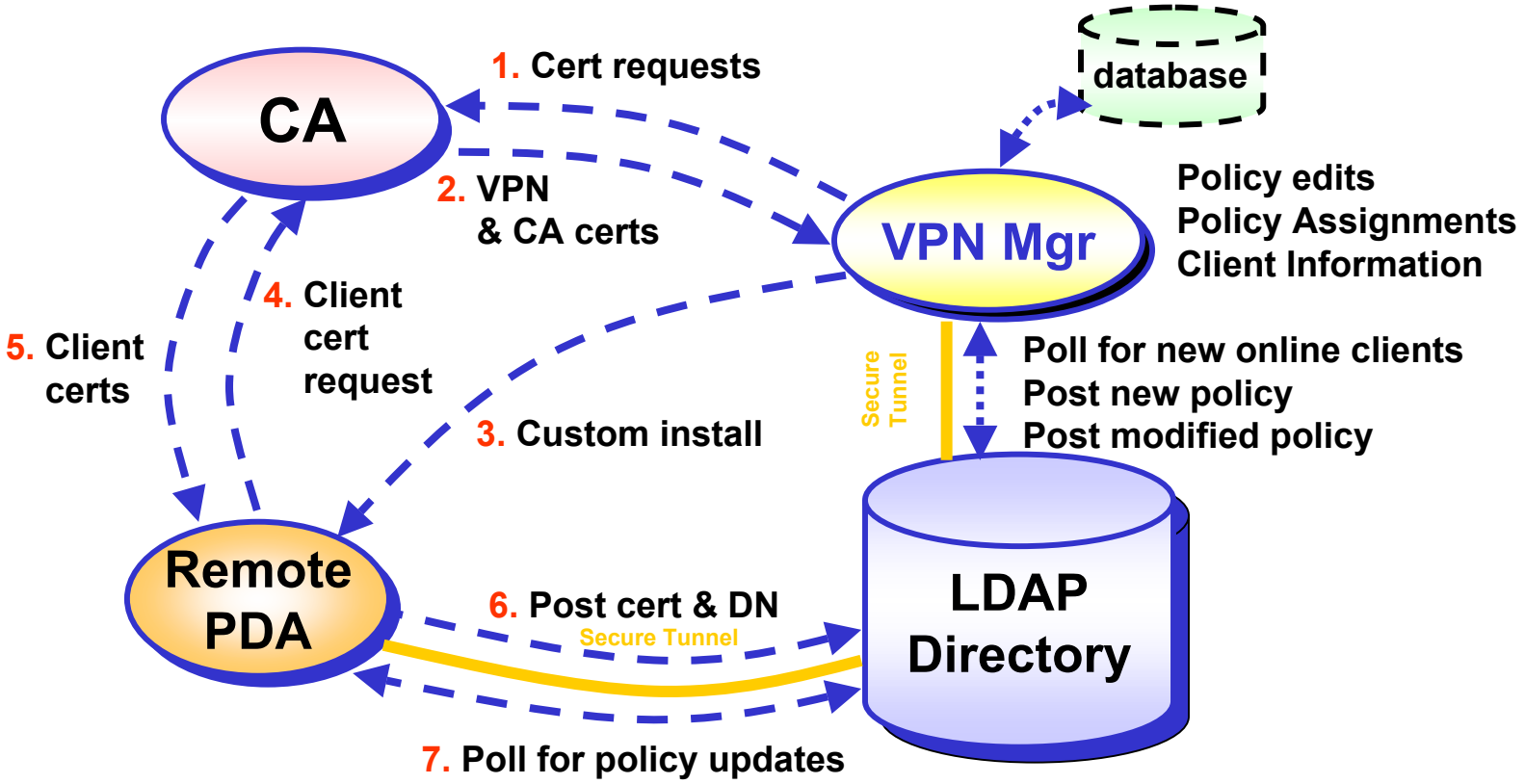
Policy

- Enforce use of solid passwords
- Utilize built-in security mechanisms
- Provide immutable forced policies
- Use standard & well-maintained anti-virus programs
- Use file encryption for internal data, folders, etc.
- Provide “device access control” – smart card, token
- Enforce use of encrypted connections, VPN tunnels!
- Provide and maintain a good firewall

Technology

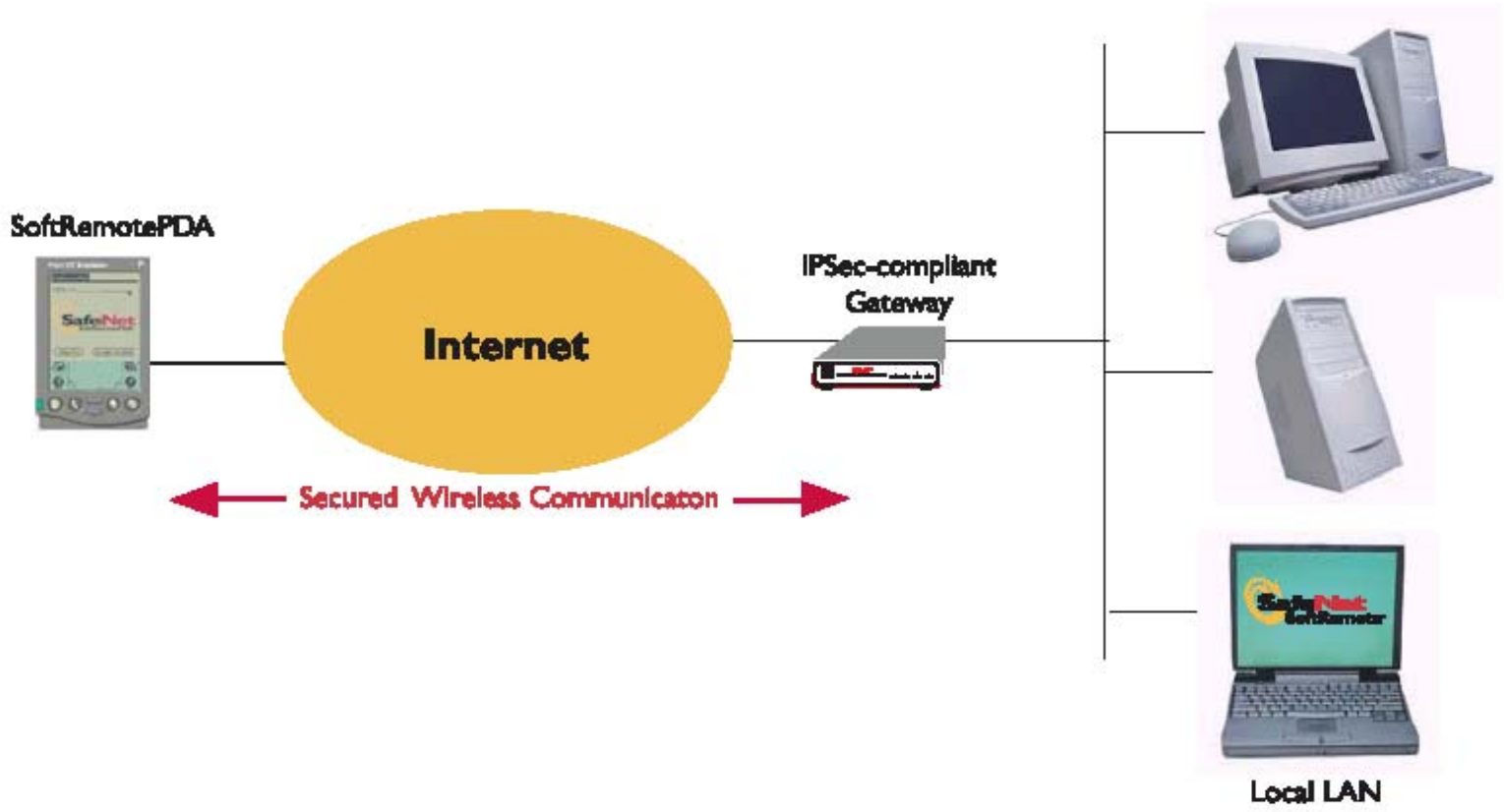
- Protocol
  - IPSec is the best, WEP, SSL
- Keys
  - 768-bit, 1024-bit most common
  - RSA signatures most common, ECC
  - Preshared secrets today, Public keys tomorrow
- Encrypting Algorithms
  - TDES most common, AES coming
- Authentication
  - 2-Factor (RSA SecurID), Certificates, PINs
  - Biometrics

Remote Policy Distribution



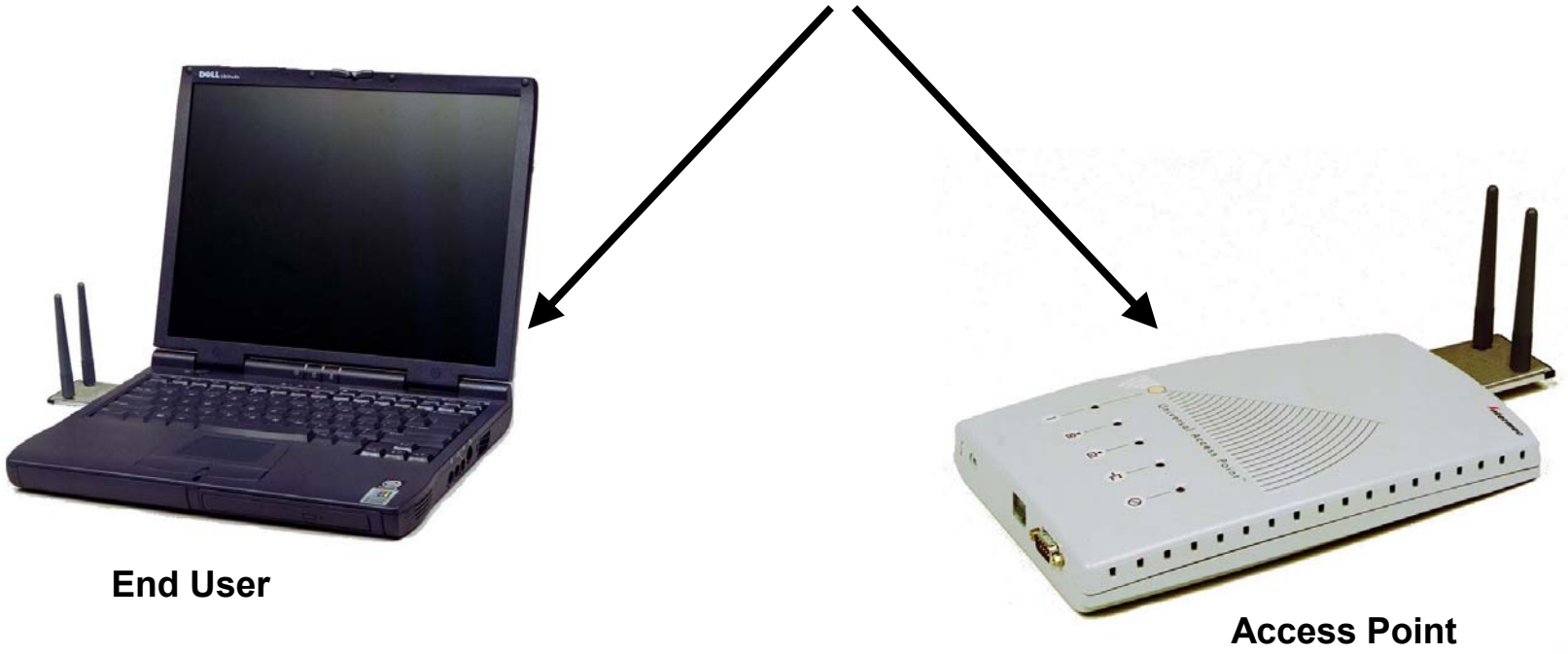
Remote PDA/Laptop user gets corporate policy …
automatically, transparently, and it's unmodifiable

PDAs/Phones



Secure <u>all communications</u> between the remote user and the corporate LAN

802.11 Wireless LAN



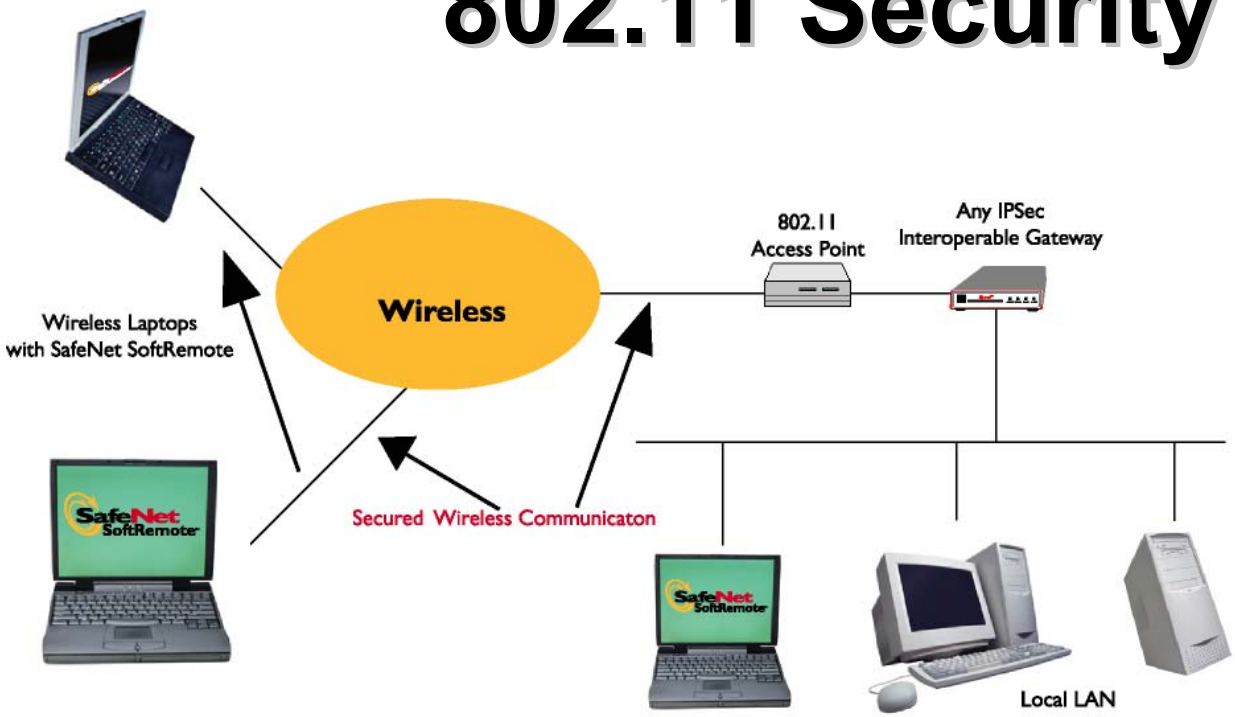
End User

Access Point

802.11b wireless LANs are likely to become the* de facto *wireless network in offices, malls, homes …

… major security vulnerability!

802.11 Security



IPSec-certified security solves 802.11 security vulnerabilities

Security concerns vanish when using IPSec security software on your laptop and an IPSec gateway behind the 802.11 access point. All information is secured with a stronger algorithm (e.g., 3DES) and at a higher IP Layer.

Procedure

What can the IT Manager do
to strengthen the corporate (remote) network?

SOFTWARE

Security in Software

HARDWARE

Security in Silicon

ACCESSORIES

Security add on devices

SOFTWARE

VPN Client

Use in PDAs, phones & laptops



- Use a VPN software client that offers secure client-to-gateway communication over wireless networks
- Use a product that is interoperable with many different gateways

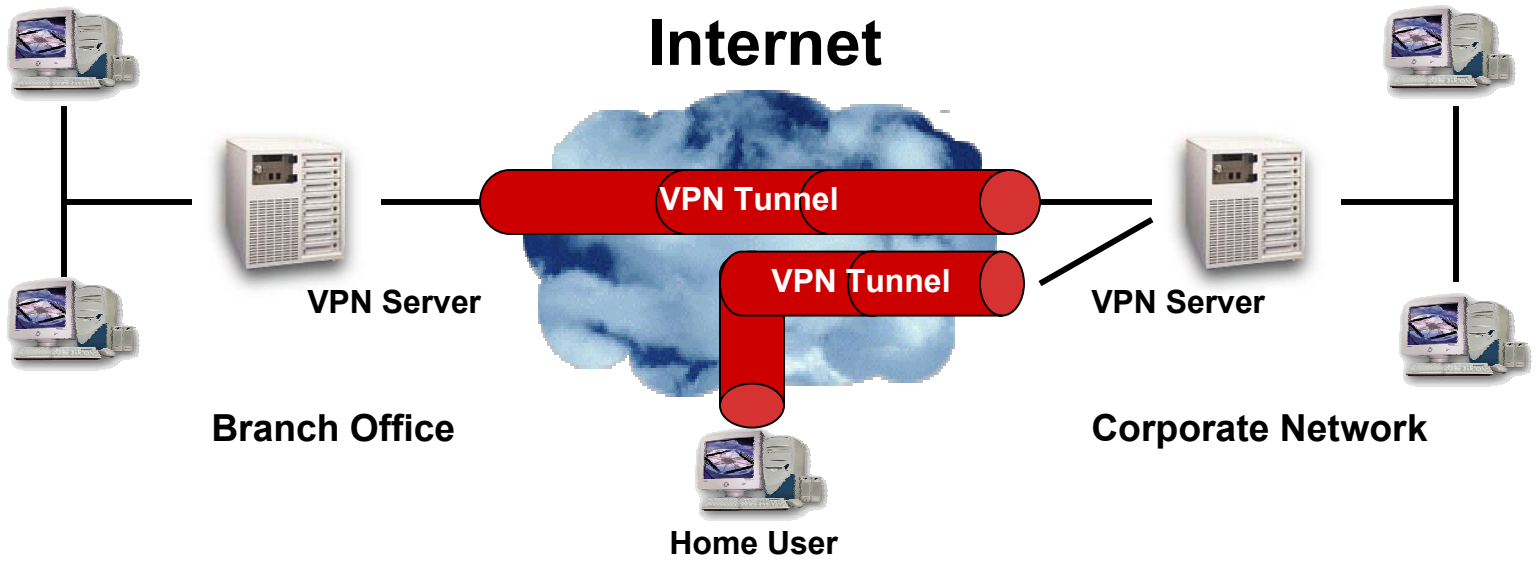






SOFTWARE

VPN: Simple, Cost-Effective



Use the Internet to establish secure links via:

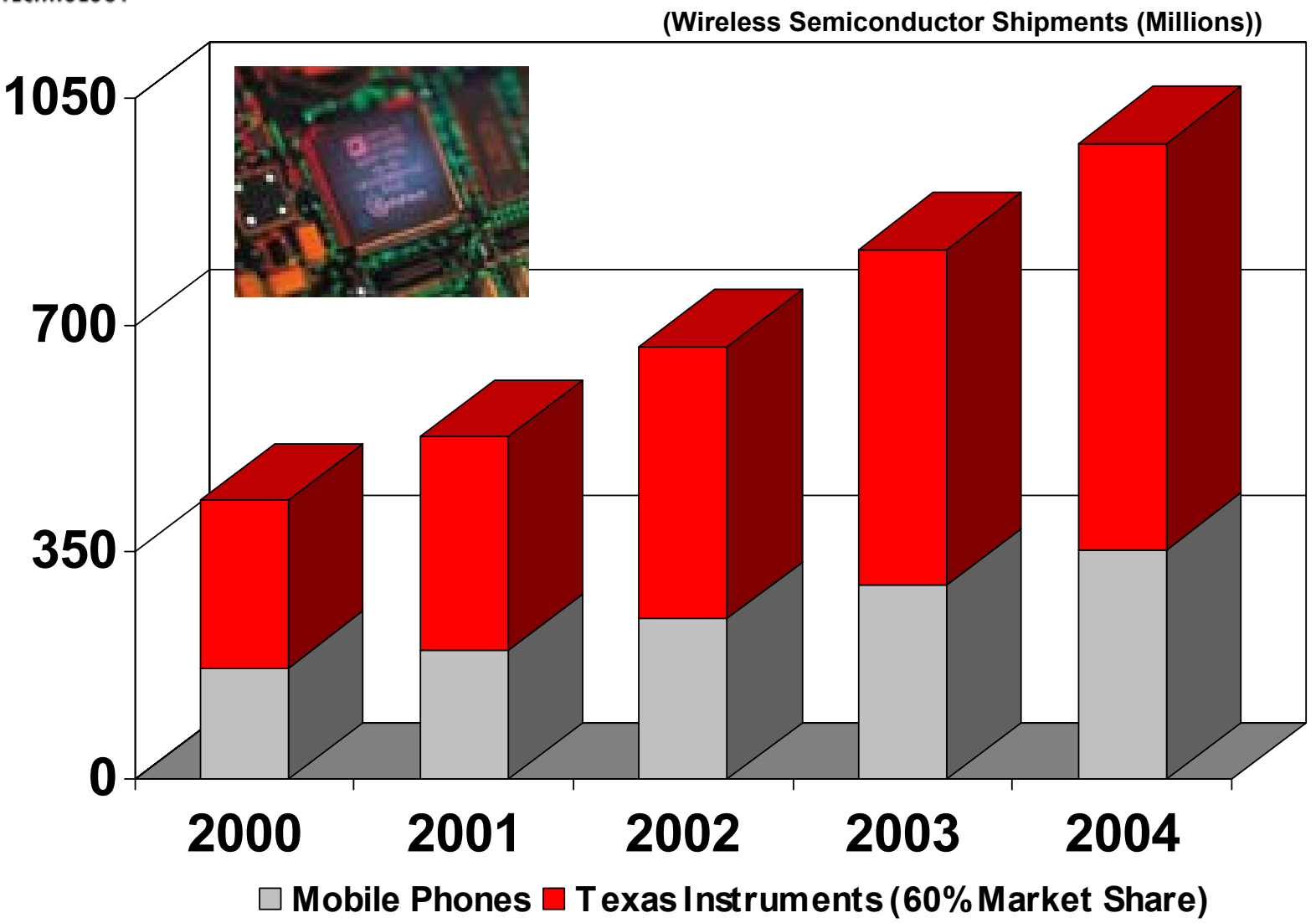
- *Authentication*
- *Encryption*
- *Secure Tunneling*

To access business partners, branch offices and telecommuters

OMAP™
TEXAS INSTRUMENTS TECHNOLOGY

HARDWARE

Hardware Security



HARDWARE

Security Embedded in Silicon

Embedded security includes all relevant cryptographic functionality in hardware form as well as a full featured software library which runs on a processor or chip.

Full functionality includes:

- **Encryption Engines: DES/3DES, ARC4, AES**
- **Hash Engines: SHA-1, MD5, RIPEMD-128/160**
- **IPSec Packet Engine**
- **Public Key Accelerator**
- **RSA, DSA Digital Signatures**
- **Entropy-based True Random Number Generator (RNG)**

2.5G and 3G device manufacturers, wireless service providers, and wireless application providers in using embedded silicon provides accessibility to all cryptographic functionality in any combination of hardware and software.

ACCESSORIES

Secure Tokens, Memory, & Device Policies

Strong, 2-factor authentication … ***built-in SecurID from RSA allows only YOU to establish secure connections***



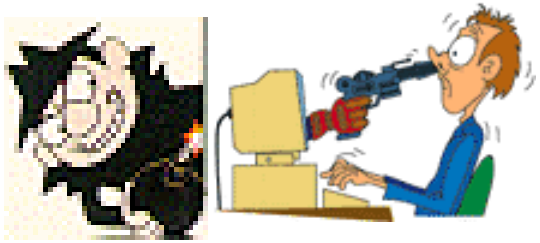
Device Access Control … allows only YOU to power up the device and/or use the select applications

Memory & file encryption products … allow the devices to be secure while unattended

Device Policy … allows specific employee access of device resources, network resources

Summary

- Assess your risk
- Protect the data on the remote phone, PDA, laptop
- Protect the network with a VPN
- Establish and enforce security policies & procedures
- Provide “remotable” client-server applications & network hardware



Give your employees access to the precious corporate resources while on the road … but do it safely, confidently and efficiently

Presented by …

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Securing Your Internet and Intranet Connection

Eric Young, Technical Director, RSA Security



An introduction to the Palm OS® SSL shared library



Agenda Slide

- SSL overview
- SSL for Palm
- SslLib API
- Example



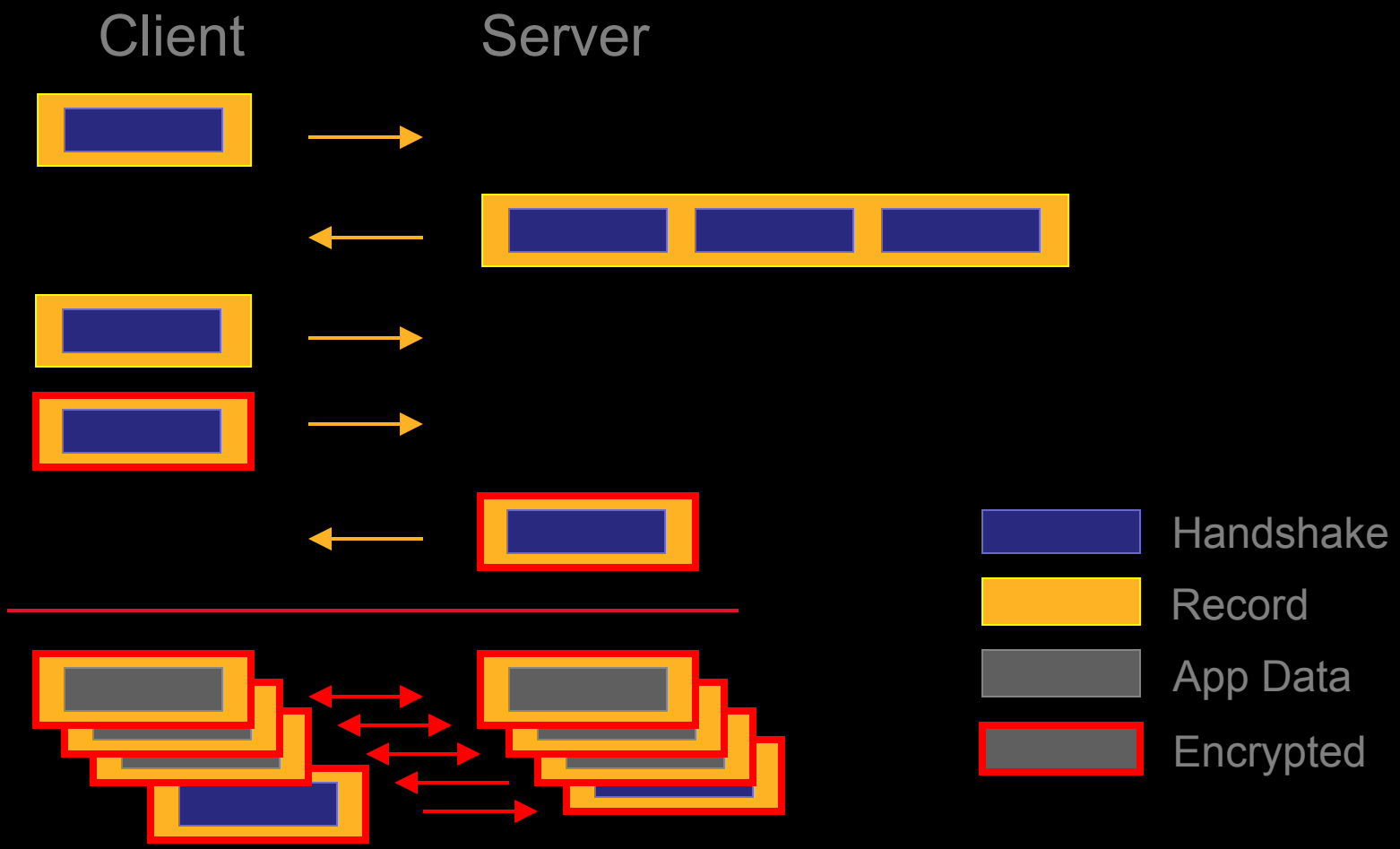
SSL Overview

- Establishes a secure connection
  - Server authentication via X.509 certificates
  - Optional Client (X.509) authentication
  - Encryption (RC4, 3DES)
  - MAC (MD5, SHA1)
- Stream protocol – not packet based
  - Similar to TCP socket API
  - Buffered read/write

SSL Protocol Overview

- Protocol exchanges SSL records
- Initial handshake exchanges messages
  - Cipher Suite
  - Shared master secret
- After the handshake the records are encrypted and MACed
- A closure alert is sent to stop the protocol
- Session reuse
  - Reuse a previous master secret
  - One less message exchange

The SSL Protocol





Palm API Design Goals

- Make it easy to use
  - Send / Receive APIs are the same as NetLib
  - Configuration against a ‘SSL Library’ structure, values inherited by individual SSL structures
- One code path for networking
  - SSL can be ‘turned off’
  - Buffered IO model
- Optimize for potentially slow networks
  - Option to read partial records
  - Option for buffered write



What is SslLib

- Shared ARM library
  - ARM native shared library
  - 68K shared library API into ARM native code
- SSLv3 protocol – client side
- RSA for key exchange and authentication
- RC4 for encryption
- MD5 and SHA1 for MAC
- Supports 128bit and export RC4 cipher suites

SslLib API Overview

- Structures
  - SslLib
    - One per application
    - Default value template
    - Information shared between SSL connections
  - SslContext
    - One per SSL connection
    - Can be used for clear-text or SSL
    - Does not create or open sockets



SslLib API Model

- Create
- Configure
- Open
- Read / Write
- Close
- Destroy

SslLib API Structure

- Create SslLib
- Set values against the SslLib
- Open a socket
- Create a SslContext
  - Set values against SslContext (including the socket)
  - Start the SSL protocol
    - Send / Receive data
  - Shutdown the SSL protocol
- Destroy the SslContext
- Close the socket, or continue using it
- Destroy SslLib



SslLib API Variants

- SslOpen is optional
  - Automatically performed on first SslSend / SslReceive
- SslClose is optional
  - Sends network messages
  - Most servers don’t require them
- A closed SslContext can be re-used
  - Session-id state is reused

Code Example

```
SslLib *lib;
SslContext *ctx;

SslLibCreate(&lib);
SslLibSet_SomeCertificateValidationOption(lib,...);
   SslContextCreate(lib,&ctx);

      // Open a socket
      SslContextSet_Socket(ctx,socket);
      SslOpen(ctx,mode,timeout); // Optional
         SslSend(ctx,buf,bufLen,flags,toAddr,...);
         SslReceive(ctx,buf,buflen,flags,fromAddr,...);
      SslClose(ctx,mode,timeout); // Optional
      // Open a socket
      SslContextSet_Socket(ctx,socket);
      SslOpen(ctx,mode,timeout); // Optional
      ...
```

Extra Things

- Session-id reuse
- Renegotiation
- Certificate validation
  - Application overrides
- Simpler SslRead / SslWrite API
  - 4 Receive / Send parameters will be automatically supplied / reused
- Buffer configuration
  - SslSend will ‘gather’ data until an explicit flush or the buffer is full
- Asynchronous / non-blocking IO
  - Careful – buffered read/write



Questions?



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